

MULTI-AGENT AI FOR GAME DEVELOPMENT

ASSIGNMENT #5

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GOAL-ORIENTED CODING AGENT

DUE DATE: August 13, 11:59 PM ET

ESTIMATED TIME: 2–3 hours

OVERVIEW

In this assignment, you will build a goal-oriented coding agent that reads a Game Design Document (GDD), scans your existing codebase, and automatically generates missing features. Your objective is to build and understand the reasoning layer; how the agent decides what to build and in what order.

There is no required language, framework, or LLM provider. You may use whatever tools and approaches work for you.

REQUIREMENTS

Your agent must be able to:

Read your GDD- Parse a game design document to extract the features and systems it describes.

Scan the codebase- Read existing source files to understand what has already been built.

Detect gaps- Compare what the GDD requires against what the codebase contains to find what's missing.

Prioritize- Decide which missing feature to build first and why.

Generate Code- Write the code for at least one missing feature.

DELIVERABLES

You must submit the following two items:

Your Goal-Oriented agent: Your complete, runnable agent. Submit the source code and any configuration needed to run it.

README

A short document explaining:

What features the agent built.

Why did the agent select that feature?

Were you able to run this in your game?